06 · Control Flow (@if, @for, @switch)

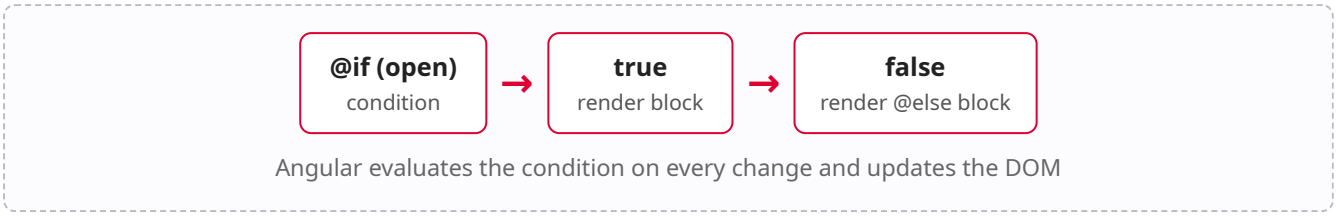
Beginner Angular learning notes

a. Definition

Control flow decides *what* appears in the template. Angular's built-in blocks are `@if` / `@else` (conditions), `@for` (repeat a list) and `@switch` (choose one of many cases). They replaced the older `*ngIf` and `*ngFor` and need no imports.

`@for` needs a **track** expression (usually an id) so Angular can reuse DOM elements efficiently when the list changes. Add `@empty` to show a message when the list is empty.

b. Diagram



Angular evaluates the condition on every change and updates the DOM

c. Syntax

```
@if (condition) { ... } @else if (other) { ... } @else { ... }

@for (item of items; track item.id; let i = $index) { ... } @empty { ... }

@switch (value) {
  @case ('a') { ... }
  @default { ... }
}
```

d. Example Program

Programs/06_Control_Flow/tea-menu.component.ts

```ts
import { Component, signal } from '@angular/core';

interface Tea {
  name: string;
  price: number;
  available: boolean;
}

@Component({
  selector: 'app-root',
  template: `
    <button (click)="open.set(!open())">{{ open() ? 'Close' : 'Open' }} shop</button>

    @if (open()) {
      <h3>Today's Menu</h3>
      <ul>
        @for (item of menu; track item.name; let i = $index) {
          <li>
            {{ i + 1 }}. {{ item.name }} - ₹{{ item.price }}
            @switch (item.available) {
```

```
              @case (true) { (available) }
              @case (false) { (sold out) }
            }
          </li>
        } @empty {
          <li>No items today</li>
        }
      </ul>
    } @else {
      <p>The shop is closed.</p>
    }
  `,
})
export class TeaMenuComponent {
  open = signal(false);
  menu: Tea[] = [
    { name: 'Filter Coffee', price: 20, available: true },
    { name: 'Masala Tea', price: 15, available: true },
    { name: 'Badam Milk', price: 30, available: false },
  ];
}
```

e. Output

Rendered in the browser (text on screen, after the UI steps below):

1. Click the **Open shop** button.

```
Close shop
Today's Menu
1. Filter Coffee - ₹20 (available)
2. Masala Tea - ₹15 (available)
3. Badam Milk - ₹30 (sold out)
```

f. Real-World Example (English + தமிழ்)

English

At a cinema counter: if tickets are available you sell, else you show 'House Full'; for every person in the queue you do the same steps; if nobody is in the queue you show 'No show'. That is `@if`, `@for`, `@empty`.

தமிழ்

ஒரு திரையரங்கு டிக்கெட் கவுண்டரைப் பாருங்கள். 'டிக்கெட் இருக்கிறதா?' என்பதே `@if`. இருந்தால் டிக்கெட் கொடுப்பார்; இல்லையென்றால் 'ஹவுஸ்ஃபுல்' பலகை காட்டுவார் (`@else`).

வரிசையில் நிற்கும் ஒவ்வொருவருக்கும் ஒரே செயலைச் செய்வது `@for`. யாருமே வரிசையில் இல்லையென்றால் 'இன்று காட்சி இல்லை' என்று காட்டுவது `@empty`. காலை, மதியம், இரவு என்று காட்சி நேரத்துக்கு ஏற்ப வேறு வேறு வேலை செய்வது `@switch`.

g. Pros

- No imports needed
- Cleaner syntax than `*ngIf` / `*ngFor`
- Faster (built into the compiler)
- `track` improves list performance

h. Cons

- `track` is mandatory in `@for`
- Older code must be migrated

i. Advanced / New Features

- Stable since v18
- `*ngIf`, `*ngFor` deprecated
- Automatic migration: `ng generate @angular/core:control-flow`

j. Common Mistakes

- Leaving out `track` in `@for` (it is required) or tracking by array index for lists that change.
- Still using `*ngIf` / `*ngFor` in new code; the new blocks are faster and simpler.
- Forgetting that signals are called like functions: `open()` not `open`.
- Putting `@else` on a new line before the closing brace of `@if` incorrectly; it must follow `}`.

k. Interview Questions

Q1. What replaced `*ngIf` and `*ngFor`?
The built-in control flow blocks `@if`, `@for` and `@switch` (stable since Angular 17).

Q2. Why is `track` required in `@for`?
It lets Angular identify items so it can move/reuse DOM nodes instead of re-creating the whole list.

Q3. What is `@empty`?
A block displayed when the `@for` collection has no items.

Q4. Do the new control flow blocks need imports?
No, they are part of the template syntax.

l. Practice Exercises

1. Show a list of 5 students; display 'Pass' or 'Fail' using `@if` based on marks.
2. Use `@switch` to show a different message for each day of the week.
3. Add a button to the tea menu that removes an item and check that `@empty` appears at the end.